11 Two stars of mass M and $9M$ are a distance d apart.

Point **P** is the point where an object will experience zero resultant gravitational force due to the two stars. The distance from the centre of mass M to P is $\boldsymbol{x}$ and the distance from the centre of mass $9M$ to P is $\boldsymbol{y}$.

What is the ratio $\boldsymbol{x} : \boldsymbol{y}$?

A 1 : 3 **B** 1 : 9 **C** 3 : 1 **D** 9 : 1